

CADET Expert System

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Introduction to CADET

- ▶ CADET: Cancer Detection Expert System
 - ▶ Rule-based medical decision support system
 - ▶ Designed for early cancer detection
 - ▶ Assists doctors, not replaces them

CADET Architecture Overview

- ▶ Inference Engine
- ▶ Knowledge Base
- ▶ Working Memory
- ▶ User Interface
- ▶ Explanation Module

Knowledge Base (Detailed)

- ▶ Hundreds of IF–THEN medical rules
 - ▶ Derived from oncologists & clinical guidelines
 - ▶ Symptoms, biopsy results, imaging findings
 - ▶ Risk factors (age, genetics, lifestyle)

Working Memory

- ▶ Stores patient-specific facts
- ▶ Updated dynamically during consultation
- ▶ Used by inference engine for reasoning

Inference Engine

- ▶ Core reasoning component
 - ▶ Mostly Forward Chaining
 - ▶ Can support Backward Chaining
 - ▶ Matches facts with rules

Forward Chaining in CADET

- ▶ Facts → Rules → Conclusion
 - ▶ Data-driven reasoning
 - ▶ Starts with patient data
 - ▶ Fires matching rules automatically

Backward Chaining (Optional Mode)

- ▶ Goal → Required Facts → Verification
 - ▶ Used for hypothesis testing
 - ▶ Efficient when checking specific cancer type

Case Study: Patient Input

- ▶ Age: 52
 - ▶ Family History: Yes
 - ▶ Breast Lump: Present
 - ▶ Biopsy: Suspicious
 - ▶ Lymph Nodes: Enlarged

Example Rule 1

- ▶ IF age > 50
 - ▶ AND breast lump present
 - ▶ THEN Moderate Cancer Risk

Example Rule 2

- ▶ IF biopsy positive
 - ▶ AND lymph nodes enlarged
 - ▶ THEN High Cancer Probability

Rule Firing Process

- Patient facts stored
 - Rule 1 conditions satisfied → Moderate risk
 - Rule 2 conditions satisfied → High probability
 - Final Output: High Risk of Breast Cancer

System Output

Diagnosis: High Risk

Confidence Level: High

Recommendation: Oncology referral

Suggested Tests: MRI, Advanced biopsy

Explanation Facility

- Explains why diagnosis was made
 - Shows which rules fired
 - Increases doctor trust
 - Enhances transparency

Flow of Reasoning

Patient Data Entered



Stored in Working Memory



Rules Matched & Fired



Risk Level Generated

Advantages

- ▶ Early detection support
 - ▶ Structured reasoning
 - ▶ Reduces diagnostic variability
 - ▶ Handles large medical knowledge

Technical Advantages

- ▶ Modular architecture
 - ▶ Easy rule addition
 - ▶ Transparent reasoning process
 - ▶ Deterministic decision logic

Limitations

- ▶ Needs frequent updating
- ▶ Dependent on rule quality
- ▶ Cannot learn automatically
- ▶ Not suitable for rare unknown cases

Real-World Deployment Challenges

- ▶ Medical validation required
 - ▶ Doctor training needed
 - ▶ Integration with hospital systems
 - ▶ Continuous knowledge updating